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Fill out this worksheet as a way to plan the data story you will make to examine your curiosity. Repeating this exercise as you work on your semester project will help ensure that you have a good data story.

1. Why did you pick your topic (i.e., problem)? That is, what made you curious about your topic (i.e., problem)?

Ans) We picked this topic because it was something we were already living every day as students on campus. A few of us work in VT Dining and have seen firsthand how overwhelming the lines get during peak meal hours. Others in our group have tried to book library study rooms last minute and found them completely unavailable days in advance. On top of that, we've all dealt with crowded buses, packed lecture halls, and slow digital systems at some point. So when it came time to choose a topic, it felt natural to look into something we were already frustrated by and curious about whether Virginia Tech's infrastructure is actually keeping up with how fast the student population is growing. We wanted to turn those everyday annoyances into something data-driven and useful, so we framed it as a "stress test" of the campus.

2. Plan and develop your data story (you will refine this as you continue your analysis, but this plan and the story you want to look for will help guide your analysis). To do so, let's first construct your outline (use the Knaflitz Chapter 7 for additional guidance):

- Your "beginning" (answer these below the question)
 - What is the setting: When and where does the story take place?

Ans) Virginia Tech's main campus in Blacksburg, Virginia, during the current academic year, as the university continues to see growth in student enrollment.



- Who is the main character: Who is driving the action?

Ans) Virginia Tech students, both, undergraduates and graduates, who rely on campus transportation, dining, classrooms, digital services, and student facilities every day.

- What is the imbalance: Why is it necessary, what has changed?

Ans) The student population at Virginia Tech has been growing steadily, but the infrastructure supporting them, including dining halls, buses, lecture halls, gyms, and digital platforms, has not scaled at the same pace. This creates daily friction that students feel but that administrators may not fully see.

- What is the balance: What do you want to see happen?

Ans) A campus where students can reliably access dining, transportation, classrooms, and facilities without significant delays, overcrowding, or frustration, and one that keeps pace with enrollment growth.

- What is the solution: How will you bring about the change?

Ans) By surfacing where the biggest pressure points are through student survey data, we can give university decision makers a clear, evidence based picture of where to prioritize investment and operational changes.

- Your “middle” (write down what you need to do for each bullet in the space provided below each bullet, you can use another sheet of paper if you need more space):

- What relevant background will you need to cover in order to further develop the situation or problem(s).

Ans) Virginia Tech's enrollment has grown significantly in recent years. Aging residence halls are under renovation, and the IT infrastructure is midway through a strategic overhaul, yet students are still experiencing daily service gaps.

- What external context or comparison points do you need to incorporate?



Ans) We can compare reported wait times, satisfaction scores, and commute challenges against stated service goals from BT Transit, VT Dining, and the VT IT Strategic Plan to show where reality diverges from what the university promises. We can also compare GrubHub wait times against non-GrubHub wait times to show whether the university's adopted mobile ordering solution is actually reducing the dining bottleneck or simply giving students a different way to experience the same problem. And we can compare library resource availability against student demand to show where physical study infrastructure is falling short of what a growing student body actually needs.

- What examples can you give that illustrate the issue?

Ans) One of us tried to grab lunch at Perry Place on a Tuesday afternoon around 12:30 PM and ended up waiting nearly 20 minutes just to get through the line, which meant rushing back to make it to a 2 PM class in Holden Hall with barely any time to eat. All of us commute from Prices Fork Road every morning and have been turned away from a full BT bus during the 8 to 9 AM as well as the 4 PM window more than once, showing up late to lecture and having to awkwardly walk in after the professor had already started. A few of us tried to book a group study room at Newman Library the day before a project deadline and found absolutely nothing available until the following week, so we ended up cramming into someone's apartment instead. And one of us sat down during registration week to update their schedule on Hokie SPA and spent a frustrating amount of time dealing with a slow, clunky interface that kept timing out, right when every minute counted. These were not rare bad days. These are things that happen to us and the people around us on a regular basis, and that is exactly what pushed us to want to study this.

- What data will you need to include to demonstrate the problem(s)?

Ans) Average dining wait times by location and time of day, GrubHub versus non-GrubHub average wait time comparison, off-campus dining willingness by class year, BT Transit satisfaction ratings broken down by commute origin, commute times and transit methods by area, time taken to reach class after arriving on campus by arrival window, classes missed due to transit or parking issues, academic infrastructure stress scores by college, foot traffic stress by building, class registration difficulties, digital service satisfaction ratings by platform, count of students wanting each platform redesigned, open ended student feedback on digital services, gym crowding patterns by time slot, library resource availability by type, and booking success rates across student service buildings.



- What actions do you think will be required as a result of your analysis?

Ans) Administrators may need to adjust dining hall staffing and hours, increase bus frequency on high traffic routes, expand study space availability, and improve the UI/UX of key digital platforms.

- What do you think may be potential options for addressing the problem(s) and how can you illustrate these?

Ans) Staggered dining hours, expanded Grab & Go options, more frequent BT routes during peak hours, better room booking systems, and prioritized upgrades to the most poorly rated digital services.

- What do you think the benefits of your recommended solution will be and how can you illustrate these?

Ans) Reduced wait times, less student stress, better academic outcomes, and a campus experience that can realistically support a larger student body.

- How can you make it clear to your audience what they should know or do as a result of your findings?

Ans) We will present visualizations that make it obvious which services are failing students most, making it easy for administrators to see exactly where action is needed.

- Your “end” (write down what you think your ending will be in the space provided below the next bullet, you can use another sheet of paper if you need more space), that is write down what you think the “call to action” will be:
 - End with a call to action: make it totally clear to your audience what you want them to do with the new understanding or knowledge that you’ve imparted to them.



Ans) We want Virginia Tech administrators, dining managers, transit planners, and IT leaders to walk away from our analysis knowing exactly which services are under the most strain and feeling equipped to act on it. Whether that means increasing dining capacity during peak hours, adding bus routes, or redesigning a digital platform, our data should make the case clear enough that doing nothing is no longer a comfortable option.



3. In the boxes below, storyboard your Tableau data story. In each box, write what that dashboard is going to be about and draw out the visual elements you will need to help tell that part of your data story. Remember, you should have a beginning, middle, and end. This is a “data story” so when you talk through this storyboard it should have a good flow.

***Visuals are submitted as a separate file named Storytelling_Q3_Visuals.**

Beginning: Who Are We Talking About? Before we make any claims, we need to show the audience who is behind the data. We start by introducing our respondents through a bar chart breaking down responses by student year, a pie chart showing gender distribution, and a horizontal bar chart showing how many students responded from each college. The story begins here because the audience needs to trust the data before they can feel the weight of what comes next.	Middle Part 1: Can Students Even Get to Campus? The story starts before the student even sets foot on campus. A horizontal bar chart shows where students are travelling from across Blacksburg and beyond, with a filter letting the audience explore which transport modes each area relies on. An area line chart tracks the average time it takes students to reach class after arriving on campus by arrival time, showing a clear spike during the morning rush. A treemap on the right shows BT Transit satisfaction scores broken down by commute origin, making it easy to see which neighborhoods feel most let down by public transit. Before we talk about dining or classrooms, we need to show that for many students the stress begins before they even arrive.	Middle Part 2: Is the Academic Side of Campus Keeping Up? Now the student is on campus and heading to class. We open with a KPI showing how many in-person classes students are taking per week to establish just how much demand is being placed on academic spaces every single day. From there we move into a lollipop chart showing how often students across each college report being blocked from taking a class they wanted due to infrastructure stress, and a treemap using color to show which buildings are absorbing the most foot traffic. The darker the building, the harder it is working, and some of them are working too hard.
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Middle Part 3: What Does a Typical Day on Campus Feel Like? The student leaves class and tries to eat. We lead with KPIs showing average meal wait time and GrubHub adoption rate so the audience immediately feels the scale of the daily grind. We then show a pie chart of GrubHub adoption, a box plot revealing how spread out meal wait times really are across all students, and a heatmap showing which dining halls are most overwhelmed and at what times of day. By the end of this part the audience should feel like they just lived a stressful lunch break themselves.	Middle Part 4: Where Are Students Actually Eating and How Bad Is It Spatially? We now zoom out and put the dining problem on a map. A geographic map of the Virginia Tech campus plots each dining hall with color coded dots showing how crowded each location is relative to the others at any selected time window, making it immediately visible which parts of campus are absorbing the most pressure. A bar chart directly compares average meal wait times between GrubHub users and non-GrubHub users, showing whether the app actually reduces the wait or just shifts it. A horizontal bar chart breaks down off-campus dining willingness by class year, revealing which student groups are most ready to give up on campus dining entirely. And a heatmap at the bottom breaks down dining location visits by student status, showing whether undergraduates and graduate students are competing for the same limited spaces.	Middle Part 5: Are the Digital Tools Keeping Up? Now we look at the tools students rely on every day outside the physical campus. A lollipop dot plot positions each digital platform by its satisfaction score making it easy to see which services are falling furthest behind. A horizontal bar chart shows which platform students most want redesigned. A KPI gives the average rating across all digital services. A stacked bar chart breaks down exactly how students rated each platform as a percentage of total responses from poor to excellent. And a word cloud built from open ended student responses puts their frustration in their own words, with terms like "outdated," "UI," "annoying," and "HokieSpa" dominating the visual.
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End: What Do Students Think and What Needs to Change? We close with the verdict. A KPI showing the overall infrastructure stress score gives the audience one number to walk away with. A second KPI shows average weekly gym visits, and a pie chart reveals which library resources students find hardest to access. A stacked bar chart shows how often students successfully book in-person appointments at service buildings, and a horizontal bar chart reveals when students think the gym is most crowded. The audience leaves knowing exactly what students are asking for and what Virginia Tech needs to do about it before enrollment growth makes all of this worse.



4. Write (in words) the story you want to tell for your project in the space that follows (write it like a real story):

Ans)

Every morning, thousands of students wake up across Blacksburg and make their way to Virginia Tech's campus. Some walk from on-campus housing, some squeeze onto a BT bus from Prices Fork Road or University City Boulevard, and some drive in from as far as Christiansburg hoping to find parking before their first class. For most of them, the day feels manageable on the surface. But underneath that routine, something is quietly breaking down.

Virginia Tech's student population has been growing steadily, and the campus they are arriving to every day was not built for this many people. We surveyed students from across the university, spanning every year from freshman to graduate and nearly every college on campus. Before we make any claims, we want to show exactly who responded and where they come from, because the story only means something if the audience trusts the people behind the data.

But the stress does not start when a student walks into a lecture hall. It starts the moment they leave home. We want to show where students are commuting from, how long it takes them to get from the point they arrive on campus to actually reaching their classroom, and how satisfied they are with the transit options available to them depending on where they live. For students coming from further out, the frustration begins before the school day even does, and that matters for everything that follows.

Once the student finally makes it to campus, we start asking harder questions about what they find when they get there. How many classes are students actually taking each week, and is the academic infrastructure holding up under that demand? We want to visualize which colleges are reporting the most friction when it comes to taking the classes they want, and which buildings are being pushed hardest by the sheer volume of students moving through them every single day. We expect the picture that emerges will show that this is not a problem isolated to one corner of campus. It is everywhere.



Then we follow that same student out of the classroom and into the rest of their day. What happens when they try to get lunch? How long are they waiting in line, and how often are they giving up and turning to GrubHub instead? We want to directly compare wait times between GrubHub users and non-users to show whether the app actually solves the bottleneck or just shifts it. Which dining halls are absorbing the most demand and at what times does that pressure become unsustainable? We want to put that problem on a map as well, so the audience can see not just when the bottleneck happens but exactly where on campus it is concentrated. And we want to show off-campus dining willingness by class year, because when students start saying they would rather leave campus entirely to eat, that is not a minor inconvenience. That is a failing system.

After the physical experience of getting to campus, sitting through classes, and trying to eat, the student then has to navigate the digital systems that are supposed to make their life easier. We want to show how satisfied students are with the platforms they rely on every day, which ones they most want to see redesigned, and what they are actually saying in their own words through open ended feedback and a word cloud that puts their frustration on full display. We expect this part of the story to reveal that the infrastructure problem is not just physical. It lives in the software too.

Finally we close with the bigger verdict. We want to capture how stressed students feel about Virginia Tech's infrastructure overall, how difficult it is to book in-person appointments at the service buildings that are supposed to support them, which library resources are hardest to get hold of, and when the gym becomes so crowded that even that outlet for stress feels out of reach. When we close on these numbers, the audience will not be able to dismiss it as our interpretation. It will be students rating, selecting, and responding in ways that tell Virginia Tech directly what is not working and exactly what needs to change before the next wave of enrollment growth makes all of it worse.



5. Now write down on this page all the questions you will need to ask of your data to confirm your story (or possibly find the real answers).

Ans)

Demographic Insights:

- Are the students who responded to our survey spread across enough year groups and colleges to make our findings credible, or are certain groups so underrepresented that they could skew what we find?
- If one college ends up dominating the responses, how do we account for that when interpreting satisfaction and stress scores across the board?

University Academia Insights:

- Is the average number of in-person classes per week consistent across all year groups or is it being driven up by one specific cohort?
- Which colleges are feeling the most academic infrastructure stress and is there a pattern where the colleges with the most respondents are actually reporting lower stress, which would be surprising?
- Which buildings are absorbing the most foot traffic and is there a connection between the most stressed buildings and the colleges that rely on them most heavily?

Dining and Daily Life:

- Is the average meal wait time consistent across all dining locations or are one or two specific halls pulling that number up on their own?
- Does switching to GrubHub actually reduce the wait time for students or are GrubHub users still experiencing waits that are comparable to those who wait in line?
- Are students who wait the longest more likely to leave campus entirely for food?
- Is the midday peak consistent across all dining locations or do some halls stay busy well into the evening while others die down?
- Is the late evening gym crowding window affecting the same students who already struggled to get food during peak hours?



Dining Locations and Access:

- Is there a relationship between where a dining hall is located on campus and how crowded students report it being, or does location not matter as much as the food options available?
- Do students of different status groups like undergraduates versus graduates tend to cluster at different dining halls or is preference spread evenly across all locations?
- Is the most popular dining hall also the one with the longest reported wait times, meaning popularity and bottleneck are directly connected?
- Which class year group is most willing to switch to off-campus dining entirely, and does that willingness increase as students progress further through their degree and become more familiar with alternatives outside campus?

Student Services & Student Experiences

- Is the average infrastructure stress score consistent across all year groups and colleges or are certain groups like graduate students or heavy campus users pulling that number up disproportionately?
- Does how often a student visits the gym affect how stressed they feel about campus infrastructure overall?
- Which library resource comes up most frequently as the hardest to access and does that vary depending on what college a student is in?
- Is there a pattern between students who report being unable to book in-person appointments at service buildings and those who rate overall infrastructure stress higher?
- Is the evening gym crowding window affecting the same students who are already dealing with dining bottlenecks earlier in the day?

Transit Insights

- Is the commute area a student travels from the biggest factor in how satisfied they are with BT Transit and does living further away consistently produce lower satisfaction scores?
- Does the mode of transport a student uses change how many classes they miss due to transit or parking issues?
- Is the morning spike in commute time to class driven by students arriving on BT buses from specific neighborhoods or is it consistent across all commute origins?
- Do students who arrive earlier in the morning report shorter times getting from campus to class or does arrival time not make a meaningful difference?



Digital Infrastructure & Web Services

- Is the platform students most want redesigned the same one that receives the lowest satisfaction ratings or are students frustrated about a different service entirely?
- Does the average rating across all digital services hold when you filter by individual college or does satisfaction vary significantly depending on which platforms a student relies on most?
- Do the words that dominate student open ended feedback align directly with the platforms that received the most votes for redesign?
- Is there a relationship between students who rate certain digital platforms poorly and those who report being unable to take a class they wanted due to infrastructure stress?